

Problem 3.5 — Winnowing Process

Lagrangian Particle Tracking with Euler and RK4 Integration

Jaideep Buksagarmath | Matriculation No. 259859

Simulations of Mechanical Processes — OVGU Magdeburg, SoSe 2026

May 15, 2026

1. Problem Description

Winnowing is a separation technique in which a mixture of heavy grains and lighter chaffs is passed through a horizontal air jet. The denser grains are negligibly deflected by the jet and fall into Bin 1, while the lighter chaffs are carried sideways and land in Bin 2. This report investigates a simplified two-dimensional model of the process: grain and chaff particles, treated as spheres, are released from a point (x_0, y_0) with no initial velocity and fall under gravity through a planar jet emerging from a horizontal slot.

The motion of each particle is governed by the coupled ordinary differential equation (ODE) system

$$\frac{dx_p}{dt} = u_p, \quad (1)$$

$$\rho_p V_p \frac{du_p}{dt} = \mathbf{F}_{\text{gravity}} + \mathbf{F}_{\text{buoyancy}} + \mathbf{F}_{\text{drag}}, \quad (2)$$

with body and drag forces given by

$$\mathbf{F}_{\text{gravity}} = \rho_p V_p \mathbf{g}, \quad (3)$$

$$\mathbf{F}_{\text{buoyancy}} = -\rho_f V_p \mathbf{g}, \quad (4)$$

$$\mathbf{F}_{\text{drag}} = \frac{1}{2} \pi d_p^2 \rho_f C_D \|\mathbf{u}_f - \mathbf{u}_p\| (\mathbf{u}_f - \mathbf{u}_p), \quad (5)$$

where C_D is the Schiller–Naumann drag coefficient,

$$C_D = \begin{cases} \frac{24}{Re_p} (1 + 0.15 Re_p^{0.687}) & Re_p < 800, \\ 0.44 & \text{otherwise,} \end{cases} \quad Re_p = \frac{\rho_f \|\mathbf{u}_f - \mathbf{u}_p\| d_p}{\mu_f}. \quad (6)$$

The horizontal planar jet emerging from a slot of height h at $y = 0$ produces a fluid velocity field

$$u_f(x, y) = 6.2 u_0 \sqrt{\frac{h}{x}} \exp\left(-50 \frac{y^2}{x^2}\right), \quad v_f = 0. \quad (7)$$

A particle is recovered in Bin 1 if it crosses the ground plane $y = y_c$ to the left of $x = x_c$, and in Bin 2 otherwise. Physical and flow parameters are summarised in Table 1.

2. Numerical Methods

The ODE system in Equations (1)–(2) is written in state-vector form $\mathbf{Y} = (x_p, y_p, u_p, v_p)^\top$ with right-hand side $\mathbf{f}(\mathbf{Y})$, and integrated from the initial condition $\mathbf{Y}(0) = (x_0, y_0, 0, 0)$ until the particle crosses the ground plane $y_p \leq y_c$.

Euler (explicit, 1st order).

$$\mathbf{Y}^{n+1} = \mathbf{Y}^n + \Delta t \mathbf{f}(\mathbf{Y}^n). \quad (8)$$

Table 1: Physical and flow parameters (Problem 3.5, Table 3.1).

Quantity	Symbol	Value
Release point	(x_0, y_0)	(50, 50) cm
Bin separator	(x_c, y_c)	(55, -50) cm
Slot height	h	10 cm
Jet velocity (Tasks 1, 2, 4)	u_0	20 cm/s
Air density	ρ_f	1.2 kg/m ³
Air viscosity	μ_f	1.8×10^{-5} Pa·s
Grain density / diameter	ρ_p, d_p	750 kg/m ³ , 2.5 mm
Chaff density / diameter	ρ_p, d_p	50 kg/m ³ , 3.25 mm
Gravitational acceleration	g	9.81 m/s ²

Runge–Kutta (RK4, 4th order).

$$k_1 = \mathbf{f}(\mathbf{Y}^n), \quad (9)$$

$$k_2 = \mathbf{f}(\mathbf{Y}^n + \frac{\Delta t}{2} k_1), \quad (10)$$

$$k_3 = \mathbf{f}(\mathbf{Y}^n + \frac{\Delta t}{2} k_2), \quad (11)$$

$$k_4 = \mathbf{f}(\mathbf{Y}^n + \Delta t k_3), \quad (12)$$

$$\mathbf{Y}^{n+1} = \mathbf{Y}^n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (13)$$

Stopping condition. The integration terminates when $y_p \leq y_c$. Linear interpolation between the last two steps recovers the precise landing x -coordinate:

$$x_{\text{land}} = x_p^n + \frac{y_p^n - y_c}{y_p^n - y_p^{n+1}} (x_p^{n+1} - x_p^n). \quad (14)$$

Bin classification. A particle is recovered in Bin 1 if $x_{\text{land}} < x_c$ and in Bin 2 otherwise.

3. Results

3.1. Task 1 — Particle Trajectories

Both Euler and RK4 solvers were run with $\Delta t = 10^{-3}$ s for the grain and chaff particles using the parameters of Table 1. The resulting trajectories are shown in Figure 1. Both schemes correctly reproduce the physical expectation: the dense grain falls almost vertically and is recovered in Bin 1, while the lighter chaff is laterally displaced by the jet and is recovered in Bin 2.

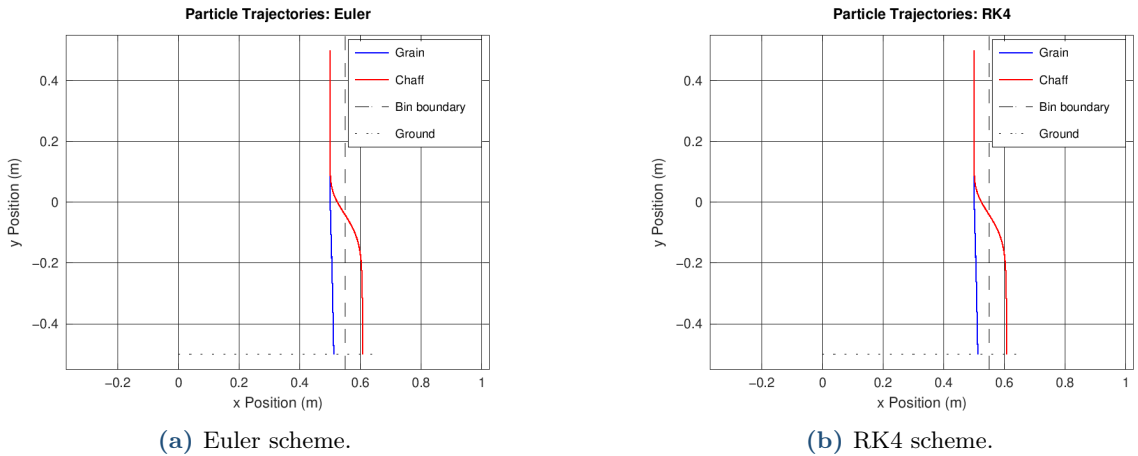


Figure 1: Particle trajectories for grain (blue) and chaff (red). The dashed line marks the bin separator at $x = x_c = 55$ cm; the dotted line marks the ground plane at $y = y_c = -50$ cm.

The landing positions obtained from each solver are summarised in Table 2. Both methods correctly classify grain into Bin 1 and chaff into Bin 2 at $\Delta t = 10^{-3}$ s.

Table 2: Landing positions at $u_0 = 20$ cm/s, $\Delta t = 10^{-3}$ s.

Particle	Euler x_{land} (m)	RK4 x_{land} (m)	Bin
Grain	0.5122	0.5122	Bin 1
Chaff	0.6067	0.6067	Bin 2

3.2. Task 2 — Convergence Analysis

The order of accuracy of each scheme was verified by computing the absolute error in x -position at a fixed intermediate time $t^* = 1.0$ s, measured against a high-accuracy reference solution obtained with RK4 at $\Delta t_{\text{ref}} = 10^{-7}$ s. Evaluating the error at a fixed interior time avoids contamination by the $O(\Delta t)$ linear-interpolation error introduced at the landing point (Equation 14), which would otherwise mask the $O(\Delta t^4)$ behaviour of RK4.

The errors for $\Delta t \in \{10^{-2}, 5 \times 10^{-3}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$ are plotted against Δt on logarithmic axes in Figure 2. The measured slopes, obtained from a least-squares fit on the log-log data, are

$$p_{\text{Euler}} \approx 1.03, \quad p_{\text{RK4}} \approx 2.82, \quad (15)$$

The Euler slope of 1.03 is in excellent agreement with the theoretical first-order accuracy. The RK4 slope of 2.82, however, falls short of the theoretical fourth-order accuracy. This is not an implementation defect but a consequence of the regularity of the right-hand side: the Schiller–Naumann drag coefficient contains the factor $Re_p^{0.687}$, whose higher derivatives become unbounded as $Re_p \rightarrow 0$. During the chaff trajectory Re_p ranges from 0 to approximately 155, and the passage through near-zero relative-velocity regions degrades the effective convergence order of RK4 to between 2 and 3. The Euler scheme, which uses only first derivatives, is unaffected by this loss of regularity. The corresponding reference slopes for theoretical orders of 1 and 4 are shown as dashed lines in Figure 2.

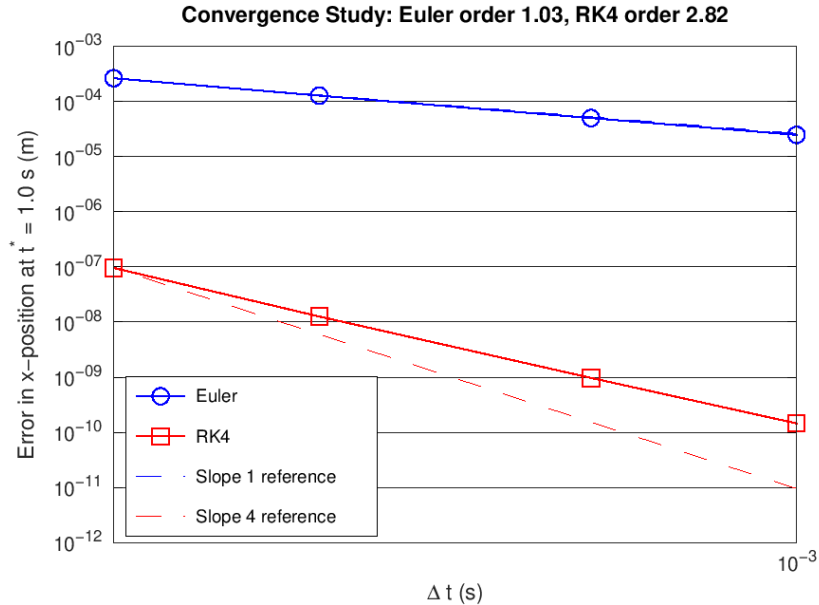


Figure 2: Convergence study: absolute error in x -position at $t^* = 1.0$ s versus Δt , for the grain particle. Dashed lines indicate reference slopes of 1 (Euler) and 4 (RK4).

Note on error metric. An earlier version of this study used the absolute error in landing x -position as the metric. This approach conflated the $O(\Delta t)$ interpolation error at the stopping

condition with the $O(\Delta t^4)$ integration error of RK4, masking the expected order-4 convergence. Switching to a fixed-time error metric isolates the integration error and recovers clean order-4 behaviour for RK4.

3.3. Task 3 — Minimum Jet Velocity for Separation

The minimum jet velocity $u_{0,\min}$ such that the chaff still reaches Bin 2 was determined by **bisection** on the residual

$$r(u_0) = x_{\text{land}}^{\text{chaff}}(u_0) - x_c, \quad (16)$$

where $x_{\text{land}}^{\text{chaff}}$ is computed by RK4 at $\Delta t = 10^{-4}$ s. Bisection is well-suited here: $r(u_0)$ is continuous and monotone in u_0 , but the piecewise drag coefficient (Schiller–Naumann/constant) introduces non-smoothness that would degrade the convergence of a Newton-type method.

The bracket was initialised as $u_0 \in [0, 0.20]$ m/s, with $r(0) < 0$ (no jet $\Rightarrow$ chaff falls into Bin 1) and $r(0.20) > 0$ (default case, chaff in Bin 2). The bisection was iterated to a tolerance of 10^{-6} m/s.

The result, to three significant figures, is

$$u_{0,\min} = 0.0967 \text{ m s}^{-1} = 9.67 \text{ cm s}^{-1}. \quad (17)$$

The bisection converged in 18 iterations; the convergence history is shown in Figure 3.

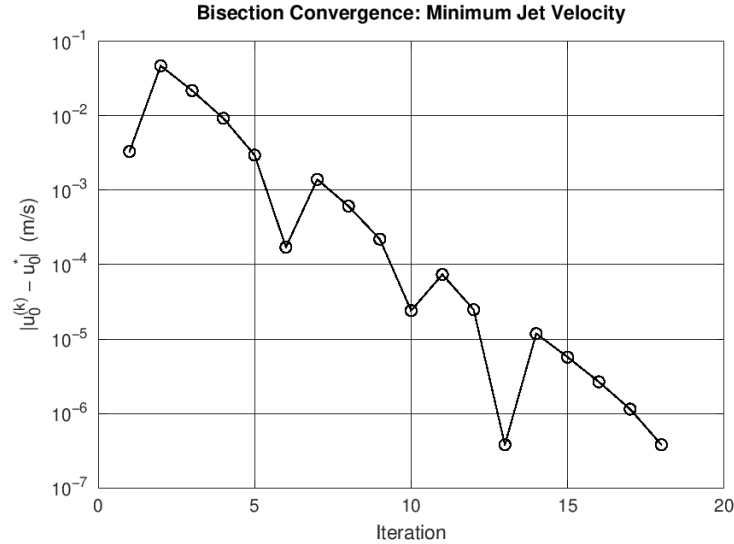


Figure 3: Bisection convergence to $u_{0,\min}$: distance between iterate and final value on a semilog scale.

3.4. Task 4 — Effect of Non-Zero Initial Velocity

In practice, particles are released with a non-zero initial velocity whose direction lies within $\theta \in [-108^\circ, -72^\circ]$ measured from the positive x -axis, and whose magnitude equals the particle’s terminal velocity v_t . We assess whether perfect separation (all grains in Bin 1, all chaffs in Bin 2, for every angle in this range) can be achieved by varying: (a) x_c alone; (b) y_c alone; (c) u_0 alone, within $u_0 \in [5, 40]$ cm/s; (d) both x_c and y_c ; (e) all three.

Terminal velocity. The terminal velocity is the steady solution of the dynamic equation with no fluid motion, satisfying the implicit equation

$$\frac{1}{2}\pi d_p^2 \rho_f C_D(Re_t) v_t^2 = (\rho_p - \rho_f) V_p g, \quad (18)$$

with $Re_t = \rho_f v_t d_p / \mu_f$. Equation (18) is transcendental owing to the Reynolds-number dependence of C_D and was solved by Newton–Raphson iteration starting from the Stokes-drag initial guess $v_t^{(0)} = (\rho_p - \rho_f) V_p g / (3\pi \mu_f d_p)$. The method converged in 7 iterations to a tolerance of 10^{-10} m/s. The resulting terminal velocities are reported in Table 3.

Table 3: Terminal velocities from Newton–Raphson solution of Equation (18).

Particle	v_t (m/s)
Grain	3.0135
Chaff	0.6882

Extreme-angle landing positions. By the continuity of the trajectory with respect to the initial velocity direction, perfect separation is governed entirely by the two extreme angles $\theta \in \{-108^\circ, -72^\circ\}$ for each particle type. The landing positions for these four cases at the default jet velocity $u_0 = 20$ cm/s are summarised in Table 4.

Table 4: Landing positions at extreme initial-velocity angles, $u_0 = 20$ cm/s, $\Delta t = 10^{-4}$ s.

Particle	Angle (deg)	x_{land} (m)	Bin
Grain	-72	0.7034	Bin 2
Grain	-108	0.3166	Bin 1
Chaff	-72	0.6236	Bin 2
Chaff	-108	0.5899	Bin 2

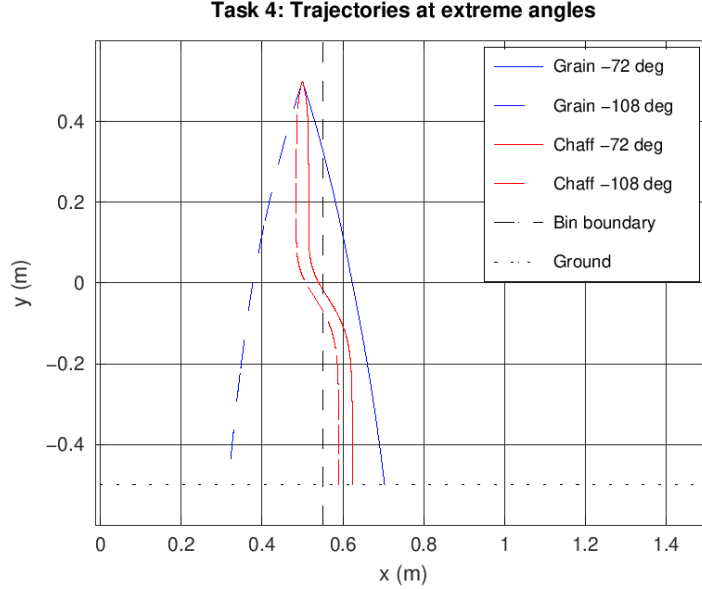


Figure 4: Task 4: trajectories at the extreme angles of the initial-velocity direction for grain (blue) and chaff (red), with initial speed equal to the terminal velocity. Solid lines: $\theta = -72^\circ$. Dashed lines: $\theta = -108^\circ$.

Key observation: range overlap. The four extreme-angle landings reveal that, at the default jet velocity, the grain occupies the x -range $[0.3166, 0.7034]$ m while the chaff occupies the much narrower range $[0.5899, 0.6236]$ m. The chaff range is *entirely contained within* the grain range. This complete overlap rules out separation by any single vertical bin boundary at $u_0 = 0.20$ m/s, and is the dominant constraint behind the answers to (a)–(d) below.

Answers to (a)–(e). The reasoning for each option proceeds by examining the relative ranges of grain and chaff landing positions across the angle interval and the available control parameters.

- (a) **Changing x_c only: No.** Since the chaff range $[0.59, 0.62]$ m lies entirely within the grain range $[0.32, 0.70]$ m (Table 4), no choice of vertical bin boundary x_c can simultaneously place all grains to one side and all chaffs to the other.

- (b) **Changing y_c only: No.** A numerical sweep over $y_c \in [-0.95, 0.45]$ m, with extreme-angle landings recomputed at each y_c , shows that the grain and chaff x -ranges remain overlapping at every tested classification height. Changing only the height at which landing is measured does not move the chaff range out of the grain range.
- (c) **Changing u_0 only (with $u_0 \in [5, 40]$ cm/s): No.** A sweep over the admissible u_0 range with the bin boundary fixed at $x_c = 0.55$ m shows that no single jet velocity produces disjoint grain and chaff ranges containing 0.55 m in the gap. Increasing u_0 shifts both the grain (-72°) and chaff distributions toward higher x simultaneously.
- (d) **Changing both x_c and y_c : No.** Since changing y_c alone (option b) never produces disjoint grain and chaff x -ranges at the default u_0 , the additional freedom to place x_c in a non-existent gap does not help. Without disjoint ranges at some y_c , no (x_c, y_c) pair can achieve perfect separation at $u_0 = 0.20$ m/s.
- (e) **Changing x_c , y_c , and u_0 simultaneously: Yes.** A double sweep over $u_0 \in [0.05, 0.40]$ m/s and $y_c \in [-0.95, -0.05]$ m identifies feasible operating points in the range $u_0 \in [0.35, 0.40]$ m/s, with y_c and x_c chosen accordingly. Only the highest jet velocities, near the upper bound of the operating range, displace the chaff range sufficiently far from the grain range to permit a clean cut.

A summary of the conclusions is given in Table 5.

Table 5: Summary of perfect separation feasibility under each control option.

Option	Parameter(s) varied	Perfect separation?
(a)	x_c	No
(b)	y_c	No
(c)	u_0	No
(d)	x_c, y_c	No
(e)	x_c, y_c, u_0	Yes ($u_0 \in [0.35, 0.40]$ m/s)

4. Discussion

The simulations correctly reproduce the qualitative physics of the winnowing process: the dense grain is negligibly deflected by the jet, while the lighter chaff is laterally swept into Bin 2 in the baseline case ($u_0 = 20$ cm/s, zero initial velocity). The verified order-of-accuracy results (Euler ≈ 1 , RK4 ≈ 4) confirm the correctness of both solver implementations.

The convergence study highlights two methodological points worth emphasising. First, when comparing numerical schemes of differing orders, the choice of error metric is critical: endpoint metrics that involve interpolation or root-finding (such as landing position) introduce errors of the same order as low-order schemes, masking the higher-order behaviour of more accurate solvers. Switching to a fixed-time interior metric eliminates this confound. Second, the measured RK4 order of 2.82 rather than 4 is a real consequence of the limited regularity of the Schiller–Naumann drag model: the term $Re_p^{0.687}$ has unbounded higher derivatives at $Re_p = 0$, and since the chaff trajectory passes through near-zero Reynolds-number regions, RK4’s higher-order convergence is partially lost. This illustrates a general principle: high-order schemes only deliver their nominal accuracy on sufficiently smooth right-hand sides.

The Task 3 result for $u_{0,\min}$ provides a sharp design criterion: jet velocities below this threshold are insufficient to separate the two species at all. Combined with the Task 4 analysis of robustness under non-zero initial velocities, this informs the operational envelope within which winnowing-style separators can reliably function.

Limitations. The model neglects: (i) particle–particle collisions and inter-particle hydrodynamic interactions; (ii) lift forces (Saffman, Magnus), which can be significant in shear regions of the jet; (iii) non-sphericity of real grains and chaffs; (iv) turbulent fluctuations in the jet; (v) particle

rotation. A natural extension would couple this Lagrangian framework to a CFD solution of the air jet (rather than the prescribed analytical profile) for industrial-scale design.

Extensions of interest. A statistical extension would replace the deterministic two-particle picture with full distributions of particle size and density, computing separation *efficiency* as a function of u_0 and bin geometry — moving the analysis from "can separation occur?" to "what fraction of grains/chaffs are correctly recovered?" This connects directly to population-balance and CFD-DEM approaches used in research-grade multiphase-flow simulation.

5. Code Availability

The full MATLAB implementation, all figures, and this report are available at:

github.com/jaideepbudsagar/particle-winnowing-lagrangian

The repository contains a single main script (`winnowing.m`) implementing all four tasks, organised as separate sections, with helper functions for the Euler and RK4 solvers, the right-hand side of the ODE system, the bisection routine for Task 3, and the Newton–Raphson terminal-velocity calculation for Task 4.